Done

: UGen : AbstractFunction : Object

Monitors another UGen to see when it is finished

Source: EnvGen.sc

See also: UGen, FreeSelfWhenDone, PauseSelfWhenDone

Description

Some UGens set a 'done' flag when they are finished playing. This UGen echoes that flag when it is set to track a particular UGen.

The UGens trackable by Done are:

- PlayBuf
- RecordBuf
- Line
- XLine
- EnvGen
- Linen
- BufRd
- BufWr
- Dbufrd
- Dbufwr
- DiskIn
- VDiskIn
- Demand

Class Methods

`Done.kr(src)`

Arguments:

src	The UGen to monitor

Inherited class methods

7 methods from UGen ► show

8 methods from Object ► show

Undocumented class methods

`Done.freeAllInGroup`

`Done.freeGroup`

`Done.freeSelf`

`Done.freeSelfAndDeepFreeNext`

`Done.freeSelfAndDeepFreePrev`

Done.**freeSelfAndFreeAllInPrev**

Done.**freeSelfAndNext**

Done.**freeSelfAndPrev**

Done.**freeSelfPauseNext**

Done.**freeSelfPausePrev**

Done.**freeSelfResumeNext**

Done.**freeSelfToHead**

Done.**freeSelfToTail**

Done.**none**

Done.**pauseSelf**

Instance Methods

Inherited instance methods

97 methods from UGen ► show

161 methods from AbstractFunction ► show

269 methods from Object ► show

Examples

The 'done' flag can be used to trigger other things in the same synth:

```
(
SynthDef("Done-help", { |out, t_trig|
    var line, a, b;

    line = Line.kr(1, 0, 1);

    a = SinOsc.ar(440, 0, 0.1*line); // sound fading out
    b = WhiteNoise.ar(Done.kr(line)*0.1); // noise starts at end of line

    Out.ar(out, Pan2.ar(a+b));
}).add;
)

Synth("Done-help"); // note that this synth doesn't have it's own doneAction, so
you'll need to manually deallocate it
```

The 'done' flag can be used to trigger a delayed freeing of the current synth, which is not possible by using doneActions alone:

```
play {
    var env = Line.kr(1, 0, 2);
    var sig = PinkNoise.ar(env);
    FreeSelf.kr(TDelay.kr(Done.kr(env), 3));
    GVerb.ar(sig, 70, 7);
}
```

Actions

can be clearer than using the number codes alone. The available doneActions are as follows:

name	value	description
none	0	do nothing when the UGen is finished
pauseSelf	1	pause the enclosing synth, but do not free it
freeSelf	2	free the enclosing synth
freeSelfAndPrev	3	free both this synth and the preceding node
freeSelfAndNext	4	free both this synth and the following node
freeSelfAndFreeAllInPrev	5	free this synth; if the preceding node is a group then do g_freeAll on it, else free it
freeSelfAndFreeAllInNext	6	free this synth; if the following node is a group then do g_freeAll on it, else free it
freeSelfToHead	7	free this synth and all preceding nodes in this group
freeSelfToTail	8	free this synth and all following nodes in this group
freeSelfPausePrev	9	free this synth and pause the preceding node
freeSelfPauseNext	10	free this synth and pause the following node
freeSelfAndDeepFreePrev	11	free this synth and if the preceding node is a group then do g_deepFree on it, else free it
freeSelfAndDeepFreeNext	12	free this synth and if the following node is a group then do g_deepFree on it, else free it
freeAllInGroup	13	free this synth and all other nodes in this group (before and after)
freeGroup	14	free the enclosing group and all nodes within it (including this synth)
freeSelfResumeNext	15	free this synth and resume the following node

For information on `freeAll` and `deepFree`, see Group and Server Command Reference.

Alternatives

Another way to free a synth when some UGen is done playing is to use FreeSelfWhenDone, or FreeSelf in combination with Done. For example, this can be used to delay the freeing to let reverb tails fade out, etc.

helpfile source: /usr/local/share/SuperCollider/HelpSource/Classes/Done.schelp
link::Classes/Done::